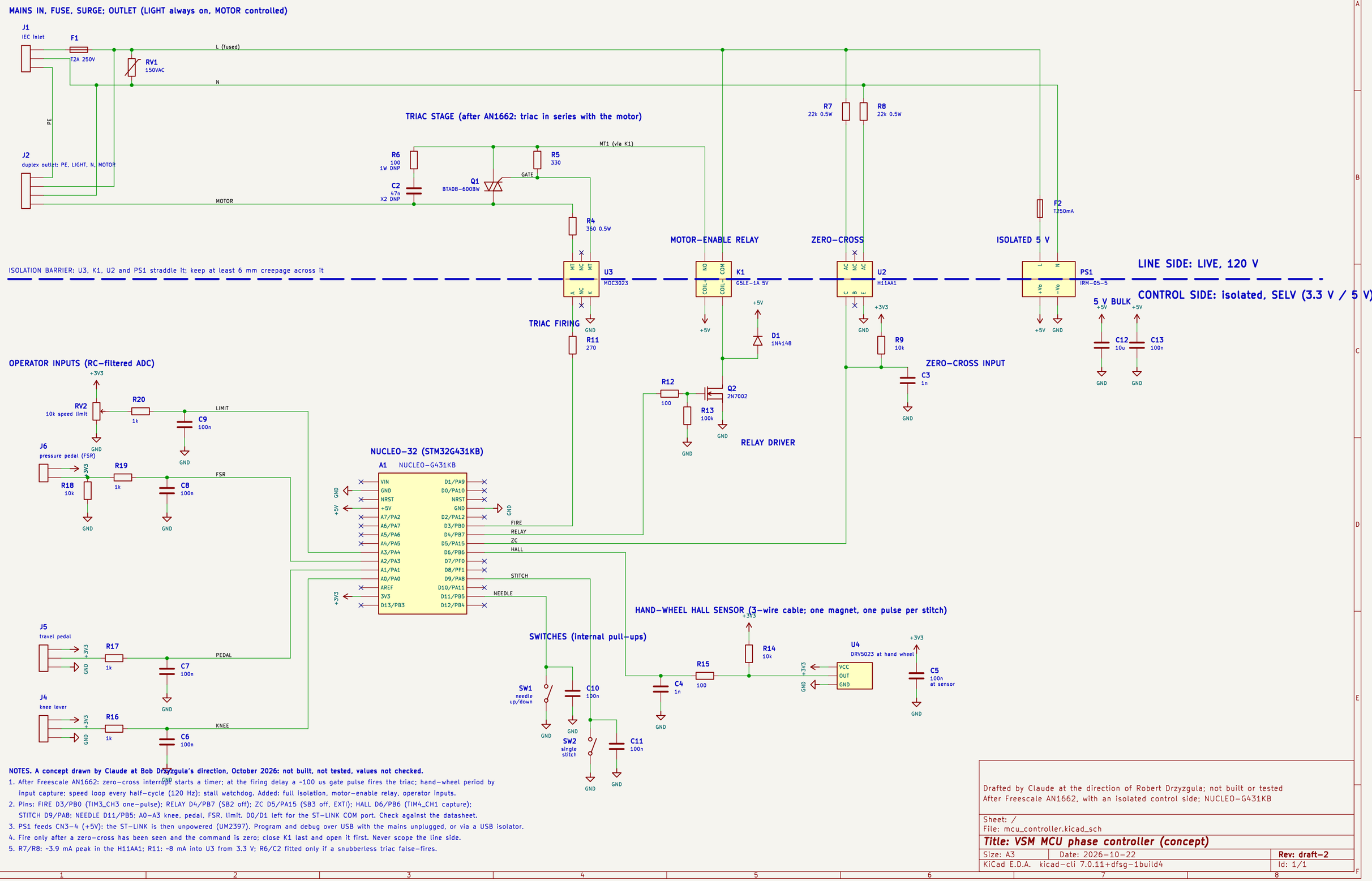

MAINS IN, FUSE, SURGE; OUTLET (LIGHT always on, MOTOR controlled)
J1
IEC inlet
F1
T2A 250V
RV1
150VAC
L (fused)
N
PE
J2
duplex outlet: PE, LIGHT, N, MOTOR
MOTOR
TRIAC STAGE (after AN1662: triac in series with the motor)
MT1 (via K1)
R6
100
1W DNP
C2
47n
X2 DNP
R5
330
Q1
BTA08-600BW
GATE
R4
360 0.5W
R7
22k 0.5W
R8
22k 0.5W
F2
T250mA
MOTOR-ENABLE RELAY
ZERO-CROSS
ISOLATED 5 V
ISOLATION BARRIER: U3, K1, U2 and PS1 straddle it; keep at least 6 mm creepage across it
LINE SIDE: LIVE, 120 V
CONTROL SIDE: isolated, SELV (3.3 V / 5 V)
U3
MOC3023
K1
G5LE-1A 5V
U2
H11AA1
PS1
IRM-05-5
5 V BULK
C12
10u
C13
100n
TRIAC FIRING
R11
270
D1
1N4148
R12
100
Q2
2N7002
R13
100k
RELAY DRIVER
R9
10k
ZERO-CROSS INPUT
C3
1n
OPERATOR INPUTS (RC-filtered ADC)
RV2
10k speed limit
R20
1k
LIMIT
C9
100n
J6
pressure pedal (FSR)
R19
1k
R18
10k
C8
100n
FSR
NUCLEO-32 (STM32G431KB)
A1
NUCLEO-G431KB
FIRE
RELAY
ZC
HALL
STITCH
NEEDLE
J5
travel pedal
R17
1k
C7
100n
PEDAL
J4
knee lever
R16
1k
C6
100n
KNEE
SWITCHES (internal pull-ups)
SW1
needle up/down
C10
100n
SW2
single stitch
C11
100n
HAND-WHEEL HALL SENSOR (3-wire cable; one magnet, one pulse per stitch)
R14
10k
R15
100
C4
1n
U4
DRV5023 at hand wheel
C5
100n
at sensor
NOTES. A concept drawn by Claude at Bob Drzyzgula's direction, October 2026: not built, not tested, values not checked.
1. After Freescale AN1662: zero-cross interrupt starts a timer; at the firing delay a ~100 us gate pulse fires the triac; hand-wheel period by input capture; speed loop every half-cycle (120 Hz); stall watchdog. Added: full isolation, motor-enable relay, operator inputs.
2. Pins: FIRE D3/PB0 (TIM3_CH3 one-pulse); RELAY D4/PB7 (SB2 off); ZC D5/PA15 (SB3 off, EXTI); HALL D6/PB6 (TIM4_CH1 capture); STITCH D9/PA8; NEEDLE D11/PB5; A0-A3 knee, pedal, FSR, limit. D0/D1 left for the ST-LINK COM port. Check against the datasheet.
3. PS1 feeds CN3-4 (+5V): the ST-LINK is then unpowered (UM2397). Program and debug over USB with the mains unplugged, or via a USB isolator.
4. Fire only after a zero-cross has been seen and the command is zero; close K1 last and open it first. Never scope the line side.
5. R7/R8: ~3.9 mA peak in the H11AA1; R11: ~8 mA into U3 from 3.3 V; R6/C2 fitted only if a snubberless triac false-fires.
Drafted by Claude at the direction of Robert Drzyzgula; not built or tested
After Freescale AN1662, with an isolated control side; NUCLEO-G431KB
Sheet: /
File: mcu_controller.kicad_sch
Title: VSM MCU phase controller (concept)
Size: A3
Date: 2026-10-22
Rev: draft-2
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4
Id: 1/1